

Lesson 15 — Design Challenge — Pattern Machine

Matches LEGO Teacher Portal lesson: "Sensory Symphony".

Objective (I can...): I can design a machine that repeats a pattern I invented using loops. **CS concept:** Applying loops and nesting to a creative build · **Builds on:** Lessons 11-14. *Also serves as flex/catch-up if the unit ran long.*

AI Connection (2 min): Repetition is how AI practices — thousands of loops to get one skill right. *First-principles question:* "Would your machine improve if it 'practiced' its pattern more, like AI does?"

Materials: LEGO CS & AI kit + device per pair (with camera), plan sheets. **Model already built** (Loops model from Lesson 11) — the build time is plan + code + test, small reconfigs only.

60-minute flow (*As the flex lesson, today's clock trims Build/Code to ~20 to make room for a 10-min AI taste. If a pair needs the full build time, run the AI taste as a whole-group demo instead.*) - **Warm-up (8):** Loop review — pairs sort example programs into "repeat N," "forever," and "nested." (*Theme: Pattern-machine showcase — a gallery walk at the end.*) - **Concept + demo (10):** Challenge: build a "pattern machine" (dancer, drummer, light show) using at least one loop. Coach models pseudocode with a loop, then hands off. (*Rotate the format: try "pair teaches pair."*) - **Build/Code (~20):** Pairs plan and code a repeating pattern that must include a loop (bonus for nesting) on the built model. *Swap at the halfway mark.* - **Meet a real AI — hands-on taste (10):** Pairs open the kit's AI/classifier feature in the Coding Canvas and show cards, small objects, or hand gestures to the device camera, watching the on-screen label change. **No training yet** — just experience a classifier reacting live. *This previews Unit 6, where they'll train their own.* **Privacy note (say it aloud):** the camera runs on-device, with no logins, and images are never saved. - **Share + pack-up (5):** Gallery walk — pairs run their pattern and point out the loop, and say one thing the classifier got right or wrong. **Photograph each build.**

Pair roles: Co-plan; Driver codes / Navigator checks the loop count; swap at 15 min. (*Trio rotates plan/code/test.*)

Differentiation: *Support* — a single-repeat target with a partial plan. *Extension / fast-finisher* — require a nested loop plus an event. *Reteach* — reuse a Lesson 12-14 loop as a template.

Exit check: Pair delivers a working pattern that uses a loop and can explain the repetition, and can say why one loop block is simpler than copying the blocks many times (**abstraction** — hiding the messy details). **Progress layer:** Send home the **Unit 3 parent note** ("your child can make a machine repeat work with loops") with the **build photo**. **At-home (optional):** Invent a light or clap pattern; write how many times each part repeats. ~10 min.